MUCOPOLYSACCHARIDOSIS TYPE II

How I did this, and why

An account of a small analysis of the IDS mutation spectrum — what I asked, how I decided, and where I was wrong before I was right

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The analysis in one paragraph

Of 395 pathogenic and likely pathogenic IDS variants catalogued in ClinVar, at most 21.5% could be corrected by a conventional base editor — and that figure is a ceiling, computed before any real-world constraint is applied. The most frequently affected codon accounts for only 2% of the annotated variants. Both findings are properties of the gene, not of any reagent.

This document is the part that does not fit inside a report. The report says what I found; this says why I asked the question at all, how I decided each thing along the way, and what I got wrong before I got it right.

I have written it plainly, because I would rather be understood than impressive. And I should say at the start that I am not writing it to ask for anything. I did the work because I wanted the answer. Writing it down carefully seemed like the least I owed it.

01

Why I chose this question

I have Hunter syndrome — mucopolysaccharidosis type II. It is caused by loss-of-function variants in IDS, the gene encoding iduronate-2-sulfatase. I have lived with it my whole life, and I have followed the genome-editing field the way other people follow a team: closely, and with more hope than is strictly reasonable.

I want to be honest about that hope, because it is the thing most likely to distort an analysis like this one. Watching the last decade of this field has been extraordinary. A disease caused by a single substitution became, within a few years, a disease with a cellular therapy behind it. Editors were built that rewrite one base without cutting the double helix. Tools arrived that can carry whole sequences into a defined site. I have read that literature as a biologist and also as someone with a personal stake in it, and I do not think the admiration is misplaced.

Over the past weeks, as part of a master's in bioinformatics, I have been learning to work with variant data. At some point I realised that I now had the tools to ask numerically a question I had only ever asked emotionally: if genome editing is going to treat MPS II one day, how much of the IDS mutation spectrum can the current generation of base editors actually reach?

I did not expect the answer to be good. I did expect it to be clear. Both turned out to be true, though less comfortably than I had imagined.

02

The question, stated narrowly

A vague question produces a vague answer, so I narrowed it until it could be counted:

> *Of the pathogenic IDS variants currently catalogued in ClinVar, what fraction could a base editor correct — even under assumptions deliberately too generous?*

Two choices in that sentence matter.

First, I restricted it to what is catalogued, not to what exists in patients. Those are different things, and the difference is the single most important limitation of this work.

Second, I chose to be generous on purpose. Had I modelled every real-world constraint, any shortfall could be waved away as my modelling being pessimistic. By computing a ceiling instead, I report a number that reality can only move downwards. If even the ceiling is low, the conclusion does not depend on my judgement.

03

Getting the data, and checking what I threw away

I retrieved IDS variants from ClinVar on 2 October 2026, filtered to germline classifications of Pathogenic or Likely pathogenic, and to variants annotated as affecting a single gene and shorter than one kilobase. That left 395.

The single-gene filter is a judgement, so I checked it rather than trusting it. I downloaded the excluded set as well — 99 variants — and looked at what was in there. Almost all of it was chromosome-scale copy-number change spanning hundreds of genes; one spanned effectively the whole of chromosome X. Those are contiguous gene syndromes, not IDS disease, and keeping them would have inflated the apparent diversity of the spectrum for entirely the wrong reason.

I kept the filter — and said in the report that I had gone and looked.

04

The first thing that went wrong

I loaded the file with pandas and printed the first few rows, which is the most ordinary thing anybody does. The Name column was showing IDS. The Protein change column was showing a disease name.

Every column was shifted by one.

I could easily have missed it. The values themselves all looked plausible — they were real data, just filed under the wrong headings — and had I gone straight to counting, I would have produced an entire analysis of nonsense without ever seeing an error message. Nothing would have crashed. The numbers would simply have been wrong.

So before anything else, I went to find out why. I opened the raw file and counted tab characters: the header row has 24 fields; every data row has 25. ClinVar's export writes a trailing tab at the end of each line, and pandas reads that extra field as a sign that the first column is an index. The fix is a single argument — index_col=False — and the whole problem disappears.

I have deliberately left both the broken load and the diagnosis in the notebook, in order. I did consider deleting them to make the work look tidier, and decided that would misrepresent how the work actually went. It also seems to me that catching this is the more useful skill, and it is invisible if the evidence is removed.

05

Deciding what each variant is

ClinVar does not give one molecular consequence per variant. It gives several, because the same change can be missense in one transcript and sit in the untranslated region of another. Counting the raw strings yields thirty categories, most of them combinations.

I resolved each variant to a single primary consequence using a severity hierarchy: frameshift, then nonsense, then splice, then in-frame indel, then start loss, then missense, then synonymous, then UTR, then intron. The reasoning is that the most disruptive annotated consequence is the one that determines what the variant does to the protein.

Then I checked whether the classification was behaving. I cross-tabulated the assigned consequence against ClinVar's own variant type and looked for combinations that ought to be impossible. No frameshift was annotated as a single-nucleotide change. No missense was annotated as an indel. That does not prove the hierarchy is the best possible one — it proves it is not contradicting itself, which is the question I could actually answer.

06

Deciding what a base editor can reach

Here the chemistry does the work, and it is simpler than it first appears.

A base editor changes what a base is, without changing how many bases there are. That single fact decides a great deal. Any insertion, deletion, duplication or microsatellite expansion alters sequence length and therefore lies outside what a base editor can do — not because today's editors are not good enough, but because of what the class of tool is. 166 of 395 variants, forty-two per cent, fall here.

For the remaining single-nucleotide substitutions, the question becomes which chemistry would be required. Adenine base editors install A to G. Cytosine base editors install C to T. Both are transitions; neither installs a transversion.

The step that is easy to get backwards is this: to correct a variant you must install the reverse of the change that caused it. A variant that arose as G to A requires an A to G correction, which is an adenine base editor. A variant that arose as A to G requires G to A, a cytosine base editor. I wrote the mapping out explicitly in the code, with a comment on every line — partly because I did not trust myself to keep it straight, and partly because anyone checking this should be able to read the reasoning rather than reconstruct it.

The result: 54 variants addressable by an adenine base editor, 31 by a cytosine base editor, and 144 that arose through transversions, for which neither chemistry applies.

PAM availability, position within the editing window and bystander editing were not modelled. All three would lower the number. That is what makes it a ceiling.

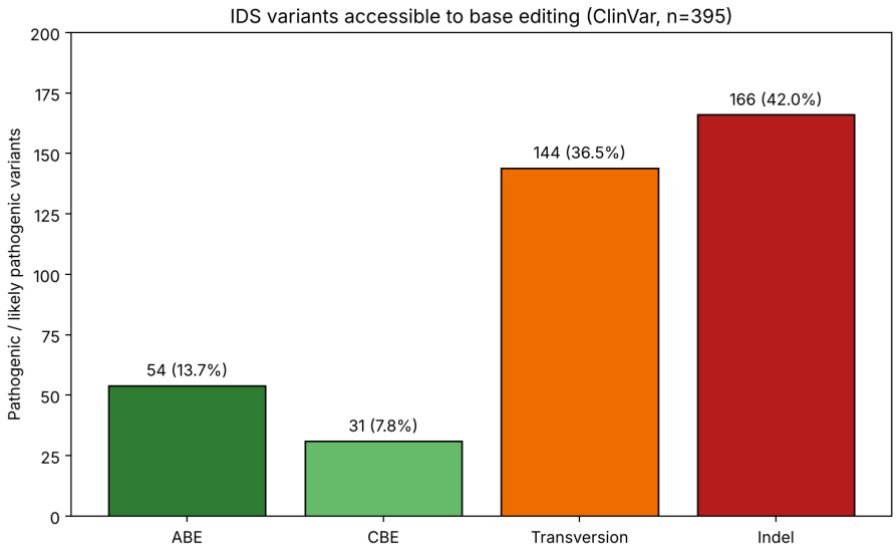


Figure 1. Accessibility of pathogenic IDS variants to base editing. ABE 54, CBE 31, transversions 144, indels 166 (n = 395).

07

The second thing that went wrong

I plotted where the variants sit along the protein and got a histogram with a wide empty channel through the middle — nothing between roughly codon 140 and codon 390. A gap that clean, in a dataset this size, is almost always a bug rather than biology.

It was. ClinVar's Protein change field lists positions for two IDS isoforms, which differ by exactly 90 residues, and it does not list them in a consistent order. One row reads G451fs, G541fs; the next reads Q531*, Q441*. My code had been taking the first number it found, which meant it was silently mixing two numbering systems and manufacturing a hole where the two ranges failed to meet.

Taking the larger value fixes it, and the range becomes a continuous 30 to 541. But I was uneasy, because “take the larger one” is a rule of thumb rather than a proof — and a rule of thumb is exactly the kind of thing that is right until it quietly is not.

So I tested it. The Name field carries ClinVar's own HGVS protein notation against the canonical transcript NM_000202.8 — for instance, NM_000202.8(IDS):c.1620dup (p.Gly541fs). I parsed the position out of that field independently and compared it, row by row, with the larger value from Protein change.

They agree for all 347 variants where both are available. Not most of them. All of them, with no exceptions.

That check took about two minutes to write, and it turned an assumption into a verified statement. It is the thing in this project I am most pleased with — and it would not exist had I been willing to let the first fix stand simply because the graph looked better afterwards.

08

The gap that was real

After the isoform fix, a smaller gap remained: nothing catalogued between c.419 and c.1180. I did not want to report it without understanding it, so I chased it.

I switched from protein coordinates to cDNA coordinates, which are isoform-independent and therefore cannot be affected by the bug I had just fixed. The gap was still there. I then looked at the variants immediately on either side of it: intronic positions, c.418+1 through c.418+3 on one side, c.1181-1 through c.1181-15 on the other. That pattern fits a region whose variants are under-reported or difficult to assign unambiguously — not an artefact of my own processing.

IDS sits beside IDS2, a pseudogene with which it recombines, and that is a well-known source of difficulty in assigning variants in this gene. I cannot prove that is the explanation, so I did not claim it. I reported the gap, said I do not believe it is biological, gave the likely reasons, and noted the direction of the bias: if variants exist there that are not captured here, the real spectrum is more diverse than my analysis shows, not less. The conclusion survives either way.

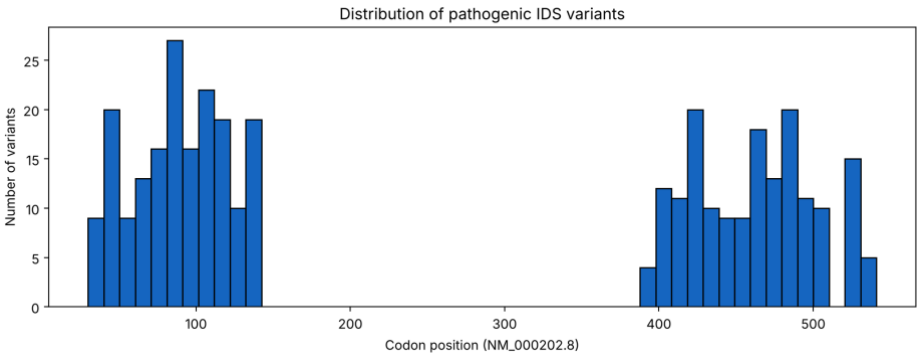


Figure 3. Distribution of pathogenic variants across the IDS coding sequence. The interval between codons ~140 and ~390 is empty in the filtered set.

09

What came out

85 of 395 variants — 21.5 per cent — are chemically correctable by a conventional base editor, judged on the nucleotide change alone and before any real-world constraint is applied. Forty-two per cent are indels, structurally out of reach of the entire class of tool. A further thirty-six per cent are transversions.

And there is no prevalent allele to target. The 347 variants carrying interpretable protein annotation fall across 167 distinct codons. The most frequently affected codon accounts for seven of them — two per cent. The five most frequent together account for nine and a half. Just over half the affected codons carry a single reported variant each.

Neither finding is about any particular reagent. Allelic diversity is a property of the gene. The restriction of base editors to transitions is a property of the mechanism. Better delivery, better specificity, higher efficiency — none of them move either number.

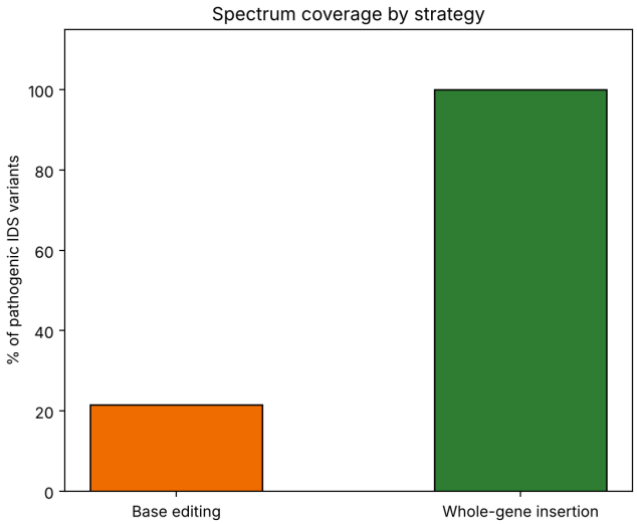


Figure 2. Spectrum coverage by strategy: per-mutation base editing (ceiling 21.5%) versus mutation-agnostic whole-gene insertion. The 100% is coverage of the reported variant spectrum — not a claim about efficacy, delivery or safety.

That places MPS II in a different regime from the diseases where per-mutation correction has gone furthest. In sickle cell disease every affected patient carries at least one copy of the sickle allele, so a single shared target serves the whole population. The approved editing therapy does not even repair that substitution — it disrupts an erythroid enhancer of BCL11A to bring fetal haemoglobin back, which helps every genotype at once. In Fanconi anaemia, correction work has concentrated on prevalent FANCA alleles because such alleles exist. For IDS there is no equivalent shortcut: a new reagent would be needed for nearly every variant, and for four fifths of the spectrum there is no base-editing reagent to design at all.

Which is, I think, the argument for approaches indifferent to which variant a patient carries — inserting a complete functional coding sequence at a defined locus, so that one reagent covers the whole spectrum. The RNA-guided bridge recombinases described recently were introduced on precisely this rationale — their authors write that such insertions would enable "correction of multiallelic genetic disorders that cannot be addressed with small-scale edits". What I have done here is put a number on one such disorder.

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What I was careful not to claim

These are counts of variants, not of people. ClinVar records what has been reported, and reporting is not prevalence. Saying how many patients a strategy would reach requires allele frequencies from a registry cohort, which I do not have and have not pretended to have. Every statement in the report is scoped to the reported spectrum.

I did not model prime editing, which reaches further — it accommodates transversions and small indels. But the edit is encoded in the pegRNA itself, so a different variant needs a different pegRNA, and efficiency varies enough with sequence context that each must be validated. Prime editing reduces the scaling problem. It does not remove it.

And 21.5 per cent is a ceiling, not an expectation. I would rather state a number reality can only lower than one I would have to defend.

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What I would do next

Three things, in order of how much they would change the picture.

First, repeat the analysis against a patient registry rather than a variant database, so that coverage refers to people rather than catalogue entries. That is the limitation that matters most, and the one I cannot fix alone.

Second, model the real constraints — PAM availability, window position, bystanders — and measure how far the ceiling falls. I expect substantially. I would rather measure it than assume it.

Third, do the same for prime editing, so that the two per-mutation approaches can be compared on the same spectrum instead of argued about in the abstract.

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A note on how this was made

I ran everything myself: downloaded the data, wrote and ran the code, hit the errors, and worked out what they were. I used an AI assistant while writing this up and while debugging, and I have said so in the repository too, because I would rather state it plainly than leave anyone wondering.

What is mine is the question, the decisions about what to count and what to exclude, the checks I chose to run, and the responsibility for anything here that turns out to be wrong. The code is deliberately plain. I wrote it to be checked, not to be admired.

I know this is a small piece of work. One dataset, one question, unsupervised and unreviewed. But it is the first time I have been able to look at my own disease and get an answer back in numbers rather than in hope — and the numbers, as it turns out, point somewhere quite specific.

That seemed worth writing down carefully, and worth sharing with people who spend their working lives on these problems. Thank you for reading it.

Data, code and figures — all 395 variants, the notebook, and everything needed to reproduce every figure — are in the accompanying repository.